

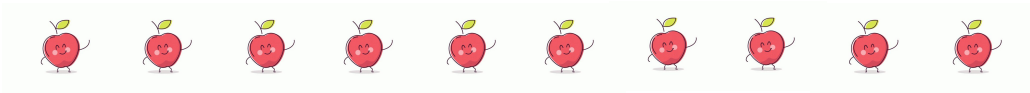
Inventory Management

Wilson Formula

- When to Order
- How Much to Order

When to Order

- $\text{Balance} < \text{Reorder Point}$
- $\text{Balance} = \text{On Hand} + \text{On Order}$
- $\text{Reorder Point} = \text{Leadtime} + \text{SS}$



Safety Stock

$$SS = z_{\alpha} \times \sqrt{E(L)\sigma_D^2 + (E(D))^2\sigma_L^2}$$

- Z alpha is the inverse distribution function of a standard normal distribution with cumulative probability alpha
- E(L) and sigma{L} are the mean and standard deviation of lead time.
- E(D) and sigma {D} are the mean and standard deviation of demand.

How Much to Order

- $\text{Order} = \text{Maximum Inventory} - \text{Balance}$
- $\text{Maximum Inventory} = \text{SS} + \text{LT} + \text{OC}$

Optimal Order Cycle

$$\text{Optimal Order Cycle (in days)} = \frac{\text{EOQ}}{\text{Average Daily Demand}}$$

EOQ

1. EOQ (Economic Order Quantity):

$$EOQ = \sqrt{\frac{2DS}{H}}$$

- D = Annual demand
- S = Ordering cost per order
- H = Annual holding cost per unit

Example 1

- On hand 12
- On Order 5
- Leadtime 7
- SS 2
- OC 1
- Balance 17
- Reorder Point 9
- Will not order

Example 2

- On hand 2
 - On Order 5
 - Leadtime 7
 - SS 2
 - OC 1
- Balance 7
 - Reorder Point 9
 - Maximum Inventory 10
- Will order
 - 3

We are not done

Factors	Unit of Measure
On Hand	Units
On Order	Units
Leadtime	Days
Safety Stock	Units / Days
Order Cycle	Days

What Demand Data to Use

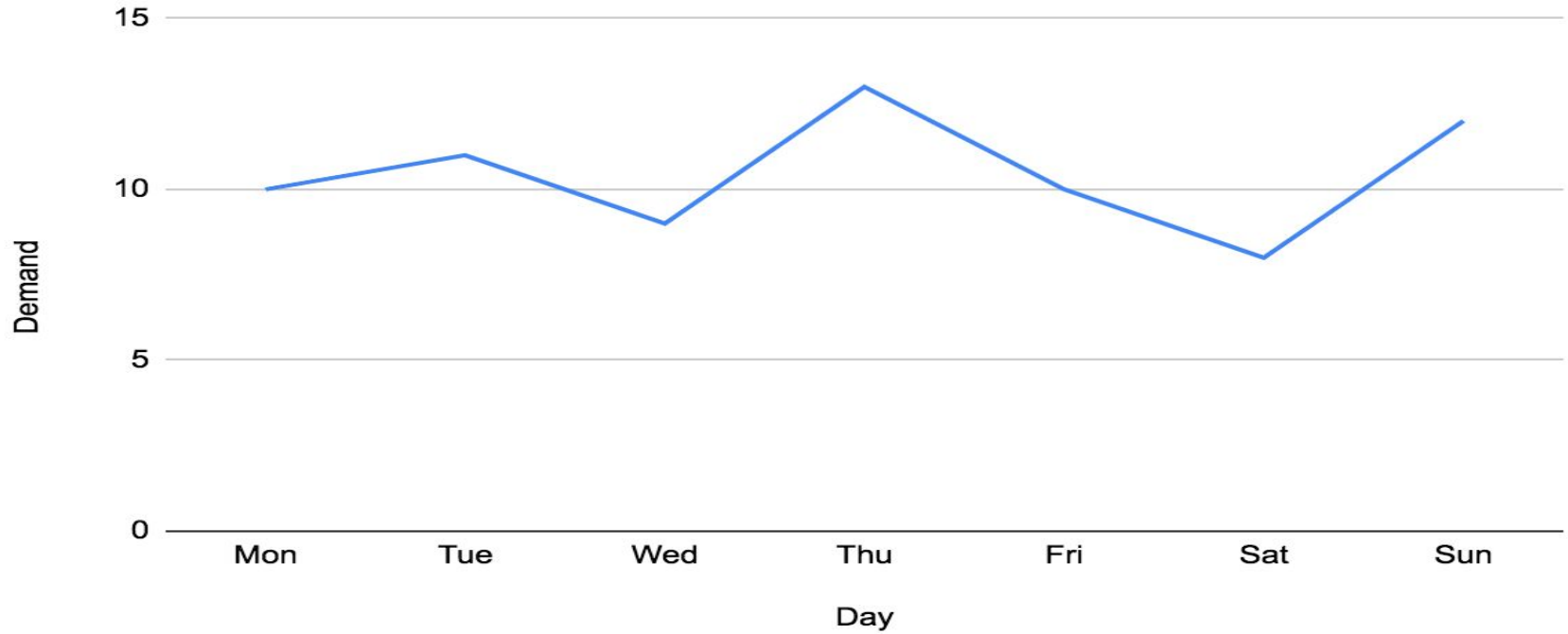
Profile of the product

Four Profiles

Profile	Data
Low MADP / High MADP	As much data as possible
Trending Up / Trending Down	Recent
Seasonal	Same season +- factor
New	Similar product +- factor

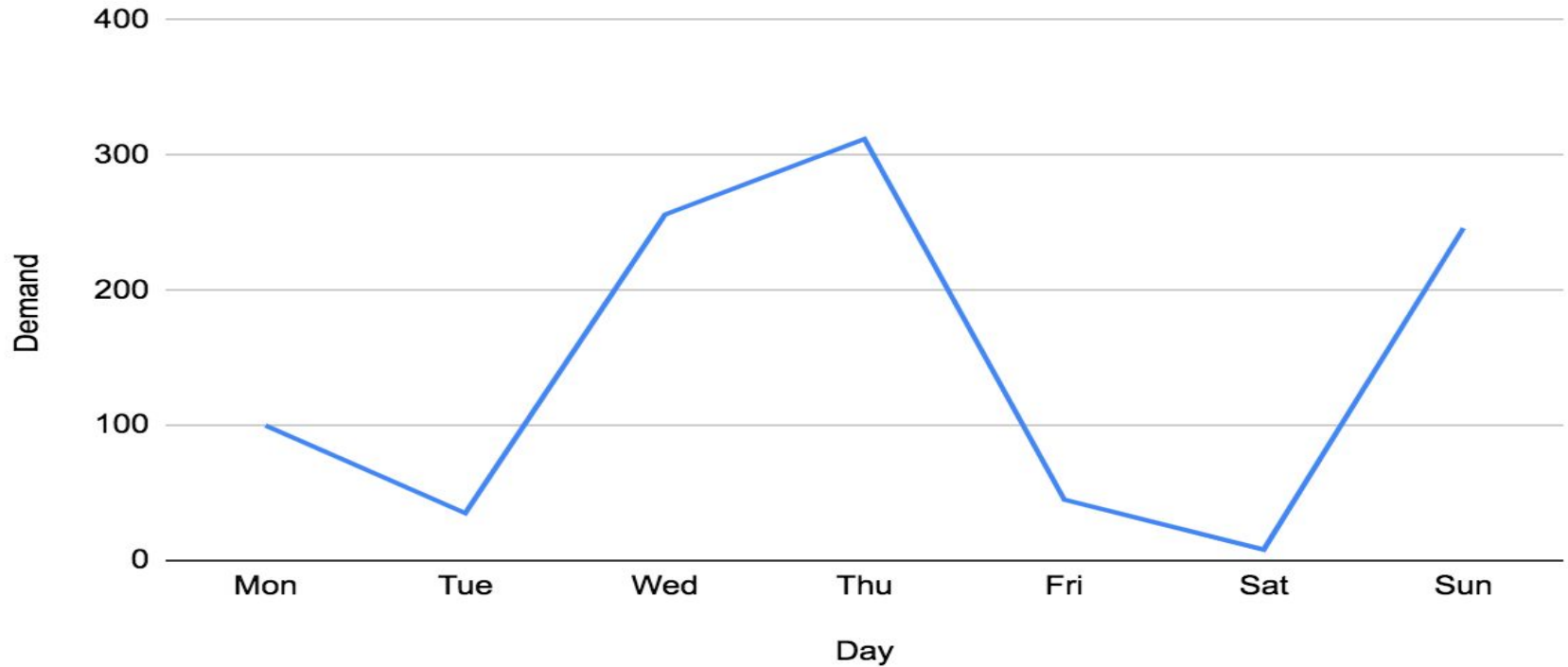
Low MADP

Demand vs. Day



High MADP

Demand vs. Day

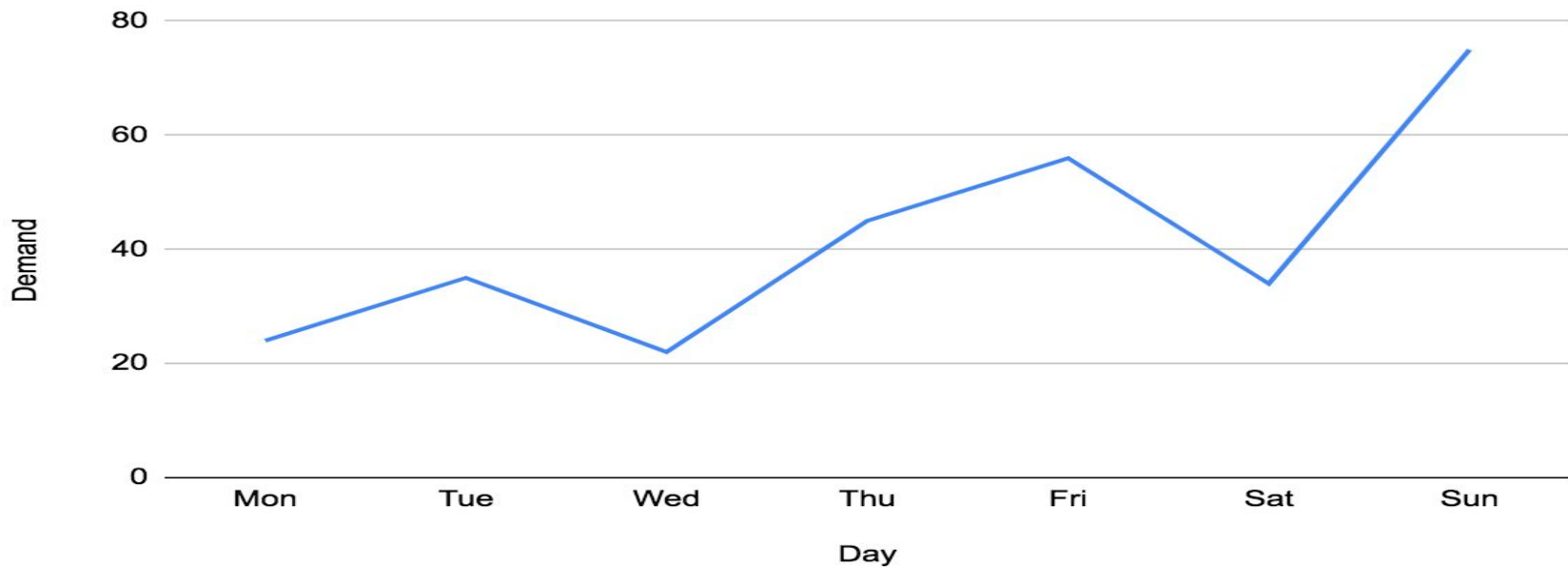


Low MADP / High MADP Data

Forecast As Much Data As Possible

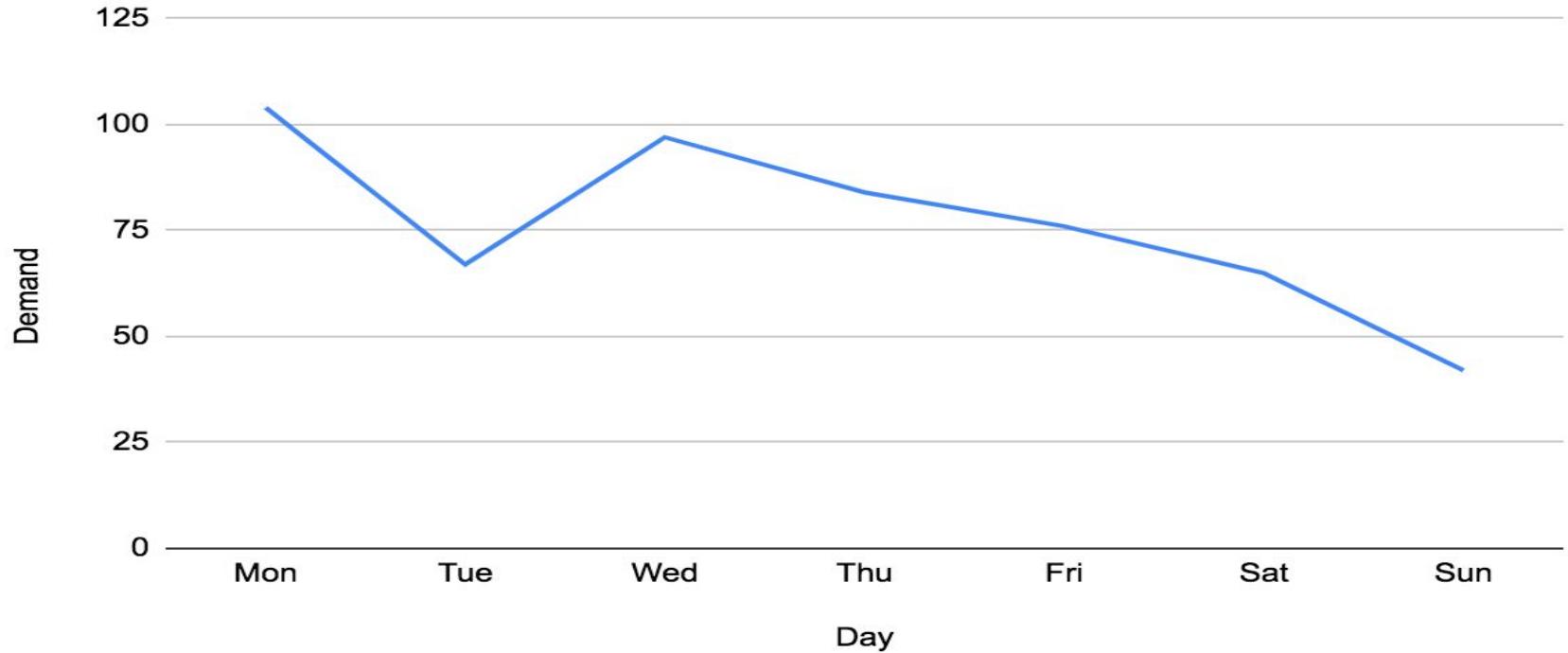
Trending Up

Demand vs. Day



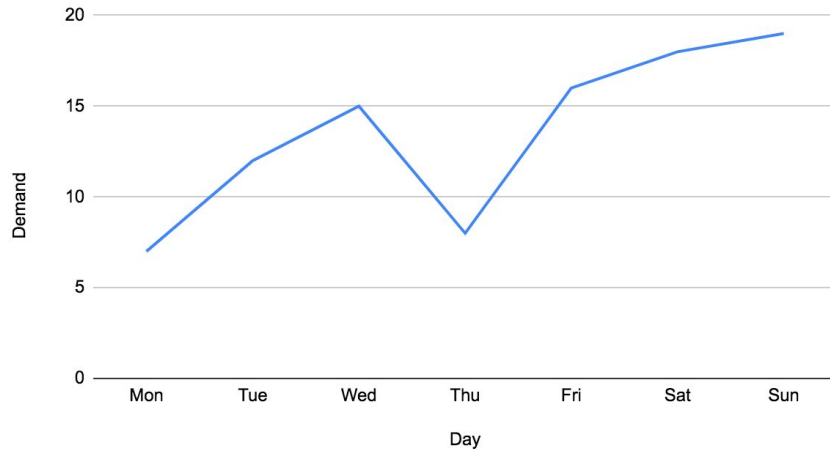
Trending Down

Demand vs. Day

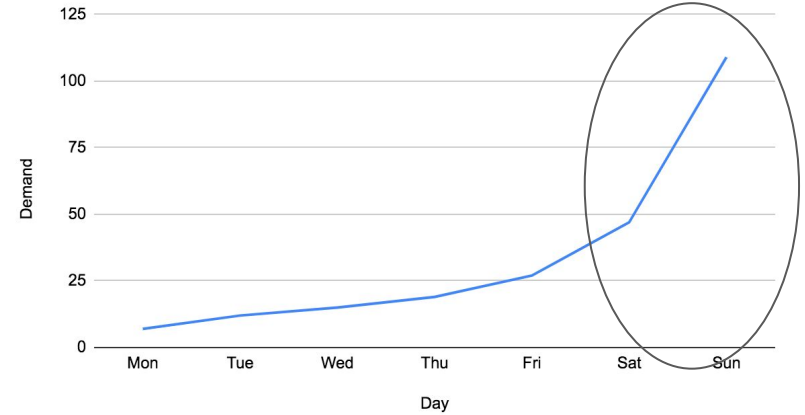


Trending Data: Recent / Trend

Demand vs. Day

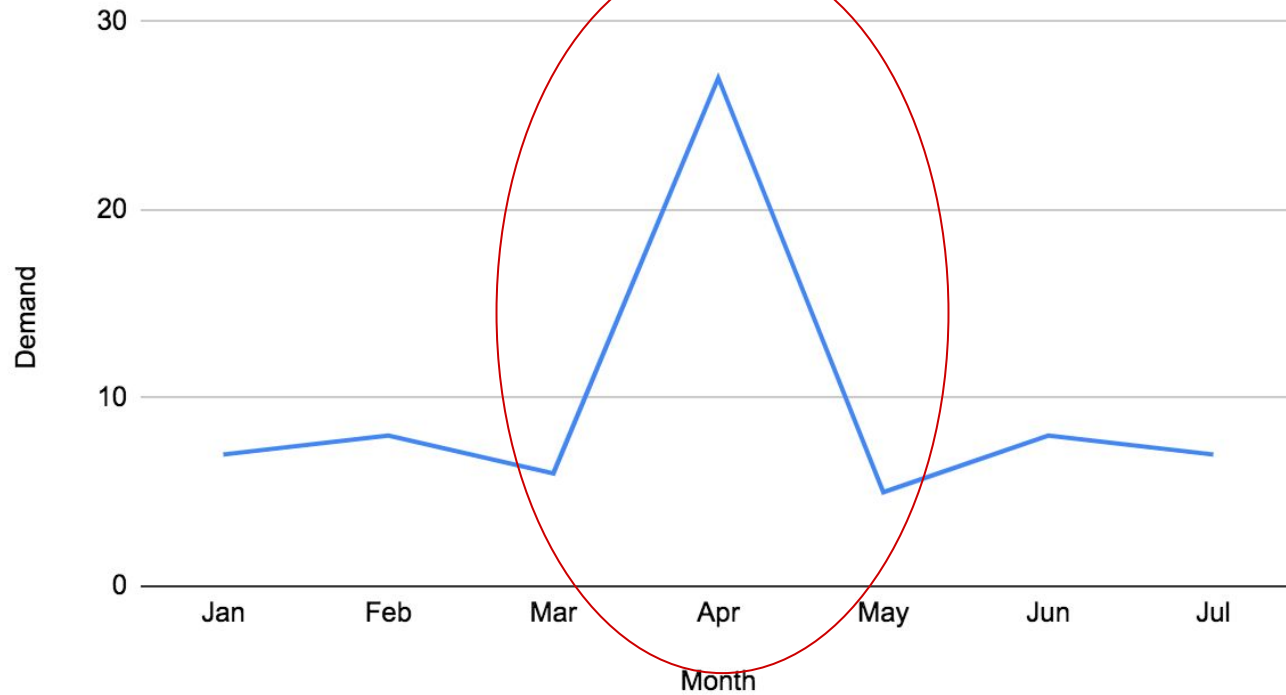


Demand vs. Day



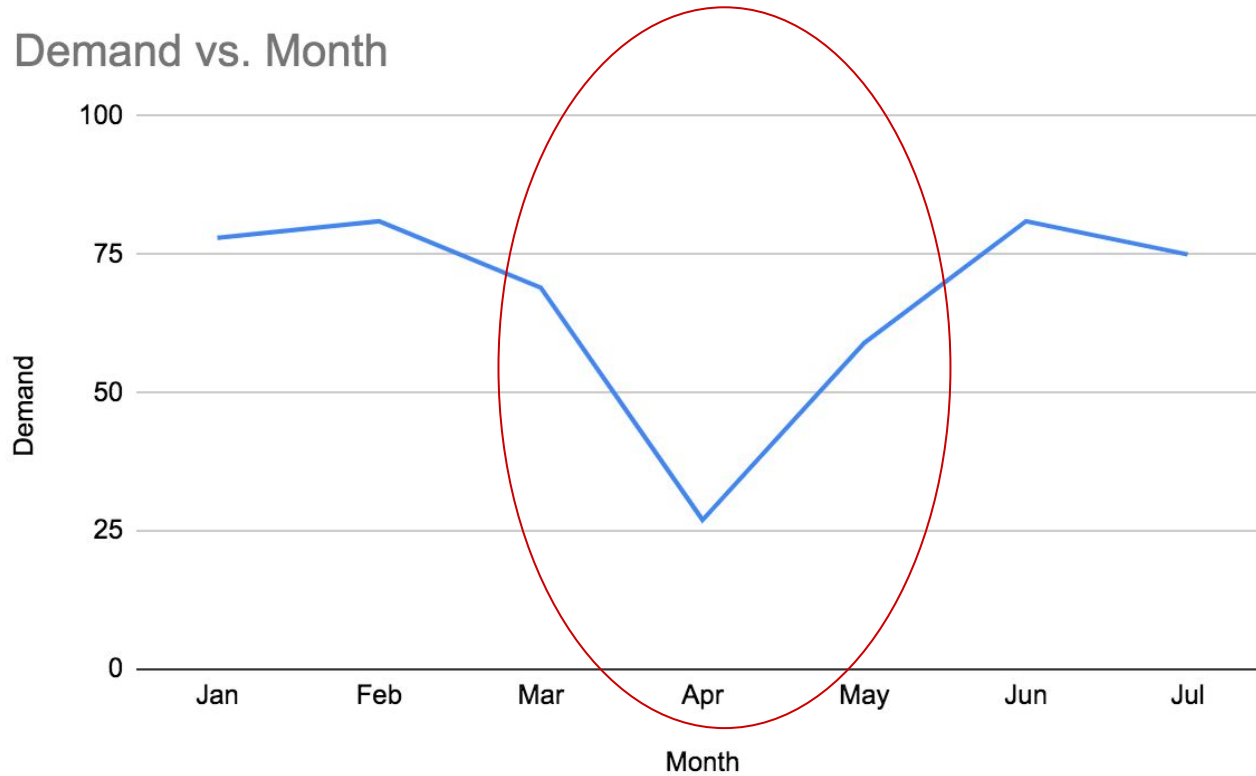
Seasonal

Demand vs. Month



Seasonal

Demand vs. Month



Seasonal

Same season +- factor

Seasonality index

New

Similar product +- factor

Market research

Adjustment

Historical + Out of Stock - Promo

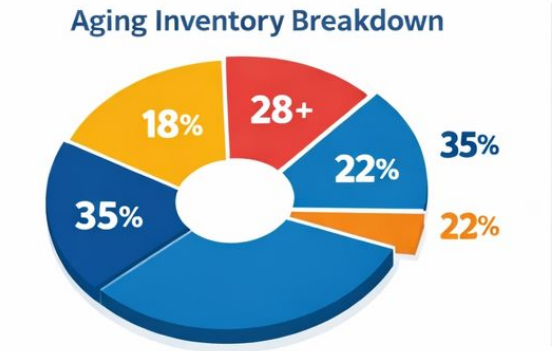
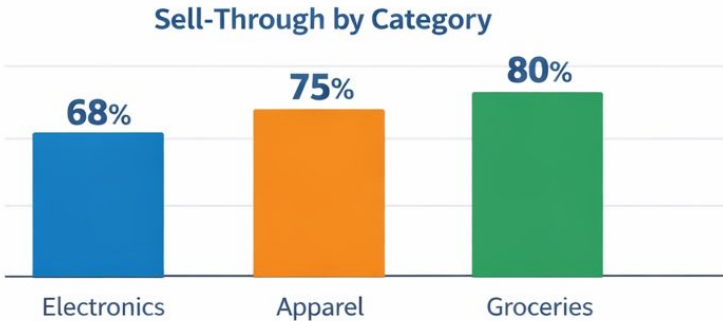
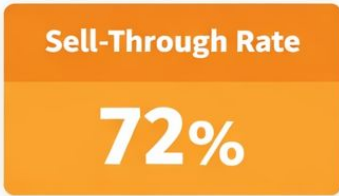
Date	Initial	Promo	Adjusted	Variance	Square
2-Jun	-	-	-	(7)	53
3-Jun	7	-	7	(0)	0
4-Jun	13	-	13	6	33
5-Jun	13	-	13	6	33
6-Jun	20	-	20	13	162
7-Jun	13	-	13	6	33
8-Jun	18	-	18	11	115
9-Jun	13	-	13	6	33
10-Jun	17	-	17	10	94
11-Jun	11	-	11	4	14
12-Jun	10	-	10	3	7
13-Jun	12	-	12	5	22
14-Jun	14	-	14	7	45
15-Jun	13	-	13	6	33
16-Jun	19	-	19	12	137
17-Jun	23	20	3	(4)	18
18-Jun	22	-	22	15	216
19-Jun	14	-	14	7	45
20-Jun	11	-	11	4	14
21-Jun	9	-	9	2	3
22-Jun	13	-	13	6	33
23-Jun	14	-	14	7	45

History $\neq$ Future

Other Factors to Consider

- Expiration / Obsolescence
- Consignment
- Case / MOQ / Pallet / Container
- Discount
- Multiple Components

Regression Analysis



At Risk Inventory

Item	On Hand	Age (Days)	Status
SKU 123	450	120	Slow Moving
SKU 456	300	95	Aging Stock
SKU 789	150	20	Healthy